Embedded System Design and Modeling

II. Modeling of discrete systems

Actor model of systems

A system can be decomposed as inter-connected building blocks, called "actors"

- Each actor has:
  - 0, 1 or more input ports
  - 0, 1 or more output ports
  - an internal computation / function / what it does
- Connections = Signals

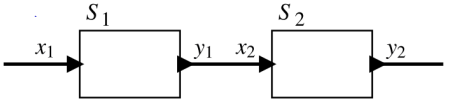


Figure 1: Actor model of systems[1]

[1](Image from Lee & Seshia (2017))

Actor dynamics

How to describe what a component does?

- Continuous dynamics (previous lecture)
- Discrete dynamics (from now on)

Ancient philosophy debate: Heraclitus (continuous) vs Parmenides (discrete)

Discrete dynamics

- **Dynamical system** = system whose state evolves in time
- **Discrete dynamics** = the system operates in a sequence of discrete steps
  - there are no continuous changes (no continuous signals)
  - like digital circuits (values change only on clock edge, e.g., rising/falling edge)
- It's more a mathematical model (real-life is continuous), but still extremely useful

Sample discrete system

Example of a discrete system model:

- Sense the cars that enter and leave a parking area (e.g., at barriers), and display the current number of cars in the parking lot on a display.

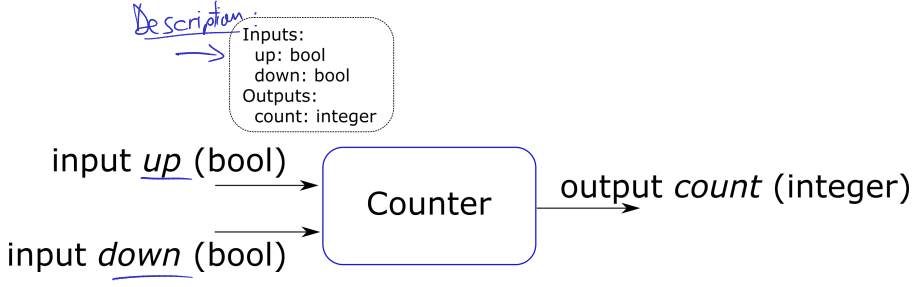


Figure 2: Parking system

State, reaction, transitions

- **State** of the system = condition of the system at a particular point in time
  - The state encompasses everything in the past that has any influence at the current moment
- When a transition's guard evaluates to true, the system **reacts**
- A **reaction** reads inputs, updates the internal state, may produce outputs, and enters a new state
- Moving from one state to the next is a **transition**.

Finite State Machine representation

- Finite State Machine = a system whose operation is described as a set of states and transitions

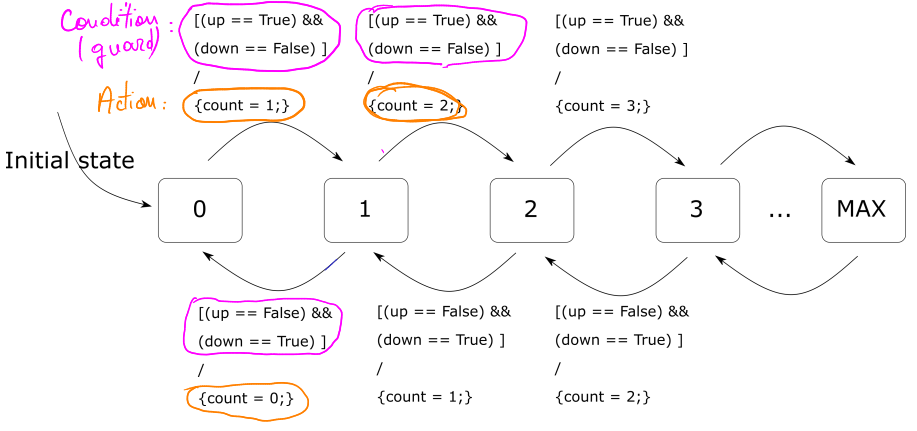


Figure 3: Parking system FSM

Components of a FSM representation

- States = the "bubbles"
- Transitions = the arrows
- Conditions (guards) = the conditions under which transitions are taken (inside "[]")
- Actions = the instructions executed when a transition is taken (after "/", inside "{}")

FSM notations

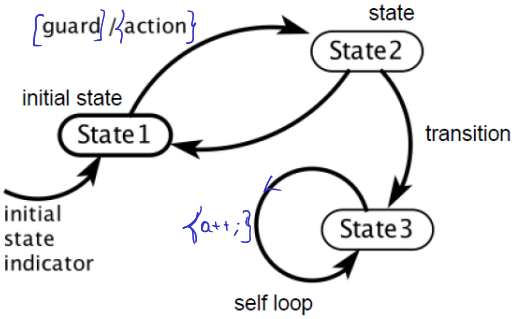


Figure 4: FSM Notations [2]

[2] Image from Seshia's slides

Conditions and actions

- A transition is taken when its guard evaluates to true
- When a transition is taken, the actions execute
- It is possible that no transition is taken, so the system preserves its state ("default transition")
- The **initial transition** indicates the starting state

FSM mathematical model

A FSM is a tuple (States, Inputs, Outputs, update, initialState) consisting of the following:

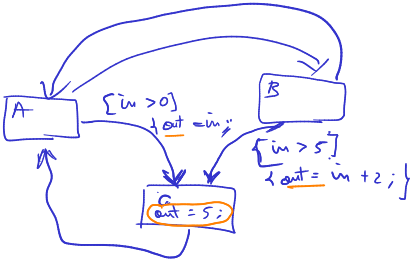
- States = a finite set $\{0, 1, \ldots, M\}$
- Inputs = a set of variables with their data types
- Outputs = a set of variables with their data types
- update = a function $f : States \times Inputs \rightarrow States \times Outputs$
  - inputs: previous state + current input values
  - outputs: next state + current output values
- initialState $\in$ States = the initial state

f(Current State, Inputs) → Next State, Outputs

} transition

If all of the above is known, everything is known about the model.

Mealy vs Moore machines



FSMs are often presented as either Mealy or Moore machines:

- Mealy machines: outputs depend on current state and current inputs
- Moore machines: outputs depend only on current state

Conditions and transitions

- Conditions and transitions can be written in many ways
- Here we use simple C-style boolean expressions:
  - == checks equality
  - ! means negation
  - `true`, `false` = boolean values
- Examples:
  - [a]
  - [!a]
  - [x >= 3]
  - [x < b]
  - etc.

a == 5
b != 0
(C)

b ~= 0
(MATLAB)

Thermostat

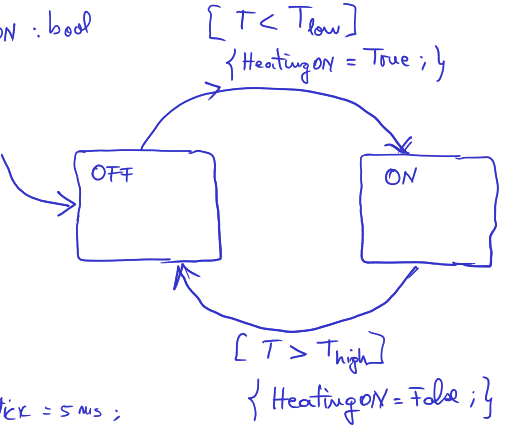
Model example: thermostat

- States: Heating OFF, Heating ON
- Guards and actions (with hysteresis $T_{low} < T_{high}$):
  - [$T \leq T_{low}$] / { heater := ON }
  - [$T \geq T_{high}$] / { heater := OFF }
- Hysteresis prevents rapid toggling around the setpoint.

19.5° C ⟶ ON

20.5° C ⟶ OFF

Inputs :
T : real
Outputs :
HeatingON : bool



When does a reaction occur?

- When are transitions checked? (when do the reactions happen)?
- Two variants:
  - **Event-triggered** model
  - **Time-triggered** model
- Event-triggered model:
  - A reaction can occur at **any time**
  - The environment triggers the transition via an **event**
  - Works like an **interrupt** in microcontrollers
- Time-triggered model:
  - The reaction occurs periodically, on the global *tick* of an **external clock**
  - e.g., everything runs at $T_s = 10$ ms, 20 ms, etc.

Time-triggered models

- Simplest case = time-triggered models
- How it works:
  - the clock ticks, the FSM "wakes up" in a certain state
  - the inputs are read
  - the outgoing transitions from the current state are evaluated
  - if a guard evaluates to true, that transition executes and the system enters a new state
  - the system "goes to sleep" until the next tick

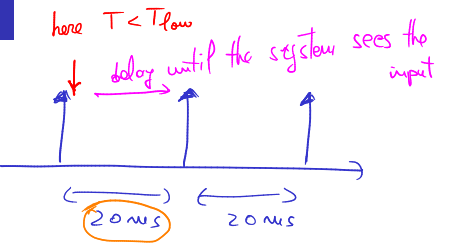
Event vs time-triggered models

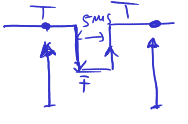
Advantages/disadvantages of time-triggered models:

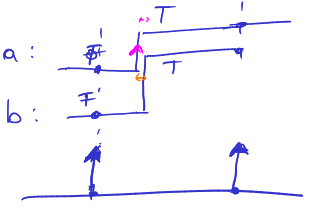
- Bad: if an input changes very fast, within a T_s interval, the model **may not see it**
- Good: all inputs are read simultaneously
- Good: simple to understand

Advantages/disadvantages of event-triggered models:

- Bad: the inputs are not synchronized (in a condition $a > b$, perhaps a changes 1 ms earlier than b, leading to a wrong result)
- Good: no risk that values are lost
- Bad: difficult to analyze, difficult to understand

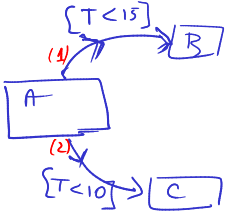




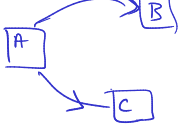


Properties of discrete models

- **Determinism**: In every state, for all possible input values, at most one transition is enabled
  - if you know the initial state and all the inputs' evolution, you know the complete behavior of the system
- **Non-determinism**: Models unknown behavior (unknown inputs), or random transitions

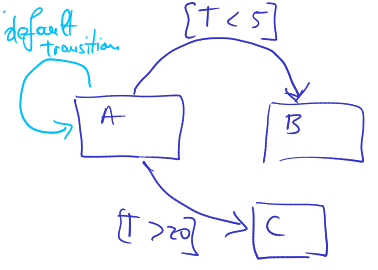


T = 7 => Which is the ~~two~~ transition which is executed?



Completeness and priorities

- **Completeness**: every input combination is handled (possibly via a default transition), so the model does not stall
- Priorities: if multiple guards can be true, define an explicit order (or disjoint guards) to keep the model deterministic



Determinism computation tree

For a fixed input sequence and initial state:

- A deterministic system exhibits a single behavior
- A non-deterministic system exhibits a set of behaviors, visualized as a **computation tree**

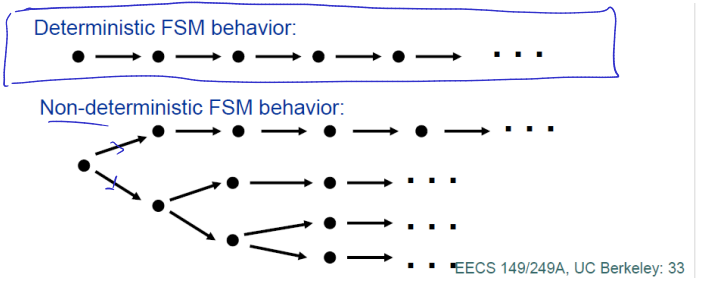


Figure 5: Computation tree [3]

[3] Image from Seshia's slides